FN-C2PA-PIOTR-001

C2PA provenance across four hops — admission pass (signed ≠ delivered ≠ instrument ≠ published)

Prepared for	Admission pass — Piotr Reder (eucomplyhub) as independent admission authority (2026-10-02)
Reconstruction ID	`FN-785efaab56d2e1f7`
Engine version	camsva 0.9.0
Generated	2026-10-02
Status	**FINAL — BOUNDED**

Purpose & boundary

This states, per proposition, what the records **establish**, **contradict**, or **cannot establish**.

ADMISSION PASS. This reconstruction runs on the observation set ADMITTED by Piotr Reder as an independent admission authority (register, 2026-10-02), within the records he can verify — the separation of authorities DEMONSTRATED, not declared. One state changed from the FN-admitted version: the LinkedIn content-match is now CANNOT ESTABLISH, because its only support (Hlias's PSNR) was not admitted and stays a declared external result. STRICTLY NON-DISPOSITIVE and instrument-agreeing. FactNotebook does NOT read the files differently from c2patool — every verdict here (Invalid / Valid / No claim found) matches c2patool 0.27.22. What it adds: (1) it binds each verdict as an ATTRIBUTED observation and states what it does and does not ESTABLISH about the artefact, across four hops — the SIGNED state (a declaration, graded as one), the DELIVERED representation (Invalid = one identified tEXt('hf-job-id') chunk vs the validating representation; removing it restores the C2PA data-hash match; the injector is CANNOT ESTABLISH), the INSTRUMENT (a 'Valid' is a verdict of c2patool under a version and trust config, not a property of the bytes — so intrinsic validity is CANNOT ESTABLISH), and the PUBLICATION hop (the LinkedIn-served bytes carry no manifest → CONTRADICTED; on the case-3 post the 'cr' badge DISPLAYS values matching the manifest while the served file carries none → displayed ≠ in-file; but what the badge reads FROM is CANNOT ESTABLISH). (2) It keeps the two CANNOT ESTABLISH reasons distinct: an evidence/relationship gap (a new observation could resolve it) vs an instrument / under-characterisation gap (the result depends on an unpinned instrument context; pinning it establishes a bounded instrument-relative proposition, not the original instrument-independent one). It never concludes 'the image is valid' or 'OpenAI produced it'. ADMISSION (per the instrument's contract): the SHA-256 recomputation (7/7) is FactNotebook's own direct read; the c2patool outputs are Piotr's runs — preserved, attributed instrument results, re-read not re-executed here. Neither is an interpretive extraction, so neither needs a separate admission step. The INTERPRETIVE candidates (README/thread/inference) were reviewed and admitted by PIOTR against his own records (OBS-C2PA-001 editorial change; 002/004/005 admitted; 003 admitted after a substantive change, final characterization confirmed by Piotr); the version divergence entered via Piotr's own reproduction (discovery Hlias); Hlias's PSNR was NOT admitted and stays a declared external result, so the content-match is CANNOT ESTABLISH. The per-candidate admission register (Piotr Reder, 2026-10-02) ships with this finding, credited both ways. No Exempt Node: FactNotebook did not admit its own interpretive candidates. Piotr Reder reviewed them against the records within his scope; FactNotebook retained responsibility for extraction, traceability and deterministic reconstruction.

Provenance & processing

An observation instrument's contract determines whether its output is a candidate observation — subjected to the applicable admission procedure before entering the reconstruction record — or a direct observation record, which does not require admission. In neither case does the instrument decide what is established. The deterministic reconstruction runs only on the admitted / authorised record set. Instrumentation may be open, heterogeneous, selective, even probabilistic; the record set is frozen; the reconstruction is deterministic.

Scope & coverage

Records were supplied by an independent source (Piotr Reder (eucomplyhub) — public C2PA fixture pack v2). Completeness beyond the supplied surface is not established. The instrumentation was selective, not exhaustive: a selective extraction is not an exhaustive observation of the surface.

See Annex C (acquisition / selection boundary) and Annex B (non-admitted candidate register).

Hop 1 — Signed state (declaration)

Proposition	State	Basis
the case-3 manifest DECLARES the claim generator 'OpenAI Media Service API' and a signing certificate (subject CN 'OpenAI Media Service', issuer 'OpenAI OpCo, LLC'), signed 2026-09-28T15:40:40Z	**ESTABLISHED**	case3 c2patool: claim_generator_info.name 'OpenAI Media Service API'; signature_info.common_name 'OpenAI Media Service'; signature_info.issuer 'OpenAI OpCo, LLC'. NB issuer = the signing certificate's issuer, not necessarily the signing entity — the two are not asserted to be the same
the case-3 manifest DECLARES the image as AI-generated (digitalSourceType trainedAlgorithmicMedia; c2pa.created by ChatGPT/gpt-image)	**ESTABLISHED**	case3 c2patool: actions.v2 c2pa.created, digitalSourceType=…/trainedAlgorithmicMedia
the case-1 SIGNED manifest declares Black Forest Labs / FLUX.2 as the generating model (claim generator 'Black Forest Labs API'; softwareAgent + modelName 'FLUX.2')	**ESTABLISHED**	case1 c2patool: claim_generator_info 'Black Forest Labs API'; modelName 'FLUX.2'; issuer 'Black Forest Labs Inc.'
the case-2 SIGNED manifest declares fal.ai as claim generator and softwareAgent, with NO model named (issuer 'fal - Features & Labels Inc.')	**ESTABLISHED**	case2 c2patool: claim_generator_info 'fal.ai'; softwareAgent 'fal.ai'; assertion label 'ai.fal.info'; no modelName field
the pack README labels case-2 'fal, Z-Image' (results table and folder name case2_higgsfield_fal_zimage) — Piotr's record of the model he selected in Higgsfield, not a statement from the signed manifest	**ESTABLISHED**	README results table + folder name 'case2_higgsfield_fal_zimage' — a SELECTION-RECORD channel (what Piotr selected in Higgsfield), distinct from the signed manifest (which names only fal.ai). ADMITTED by Piotr (editorial change; subject, channel, grade and scope unchanged)

Hop 2 — Delivered / representation

Proposition	State	Basis

the delivered PNG (cases 1–2) passes the C2PA data-hash integrity check (a valid C2PA artefact UNDER that check)	**CONTRADICTED**	CONTRADICTED by the C2PA data-hash check on the delivered bytes (assertion.dataHash.mismatch → Invalid) — an instrument-independent hash comparison. Scoped to that check, not an absolute 'valid file' property (Hop 3 holds the version/trust half of 'Valid' separately)
the delivered PNG differs from the validating representation by one identified tEXt('hf-job-id') chunk — removing that chunk restores the c2pa DATA-HASH MATCH	**ESTABLISHED**	chunk_order.txt: delivered = IHDR tEXt(hf-job-id) caBX IDAT… IEND vs removed = IHDR caBX IDAT… IEND; after removal the c2pa.hash.data matches (instrument-independent). The full 'Valid' verdict ALSO depends on the certificate/trust half (config-dependent — Hop 3); cases 1–2 are Valid on both core 0.90 and 0.91 (nothing further isolated). 'Content' here = bytes covered by the data hash (a tEXt chunk), not the rendered pixels; this does not establish who produced or signed the underlying image

Hop 3 — Instrument

Proposition	State	Basis
c2patool 0.27.22 with no trust list returned Valid for case 3 (signingCredential.untrusted only)	**ESTABLISHED**	case3 c2patool.txt: validation_state 'Valid'; only failure code signingCredential.untrusted
the case-3 validity verdict DEPENDS on validator version + trust config (c2pa-core 0.90.0 Valid; 0.91.0 Invalid 'certificate missing required EKU'; Valid again with an OpenAI EKU OID allow-listed)	**ESTABLISHED**	DISCOVERY stays attributed to Hlias Staurou. ADMITTED here via Piotr's OWN macOS reproduction (c2pa-python 0.37 & 0.38, core 0.90 & 0.91), case 3 v2 — his v1 run matched Hlias's log line-for-line. Which certificate in the chain 0.91 rejects on (leaf / intermediate / TSA) was NOT isolated; no judgment on whether the 0.91 change is correct. Cases 1–2: Valid on both versions

Hop 4 — Publication hop (LinkedIn)

Proposition	State	Basis
the LinkedIn-served image bytes — the six size variants inspected 2026-09-29 — carry an in-file C2PA manifest	**CONTRADICTED**	CONTRADICTED by Piotr's in-page inspection of six served size variants (2026-09-29, 20–1280px): progressive JPEG, JFIF only, no APP11/JUMBF segment, no OpenAI reference. (Piotr did NOT record a 'c2pa' string search; the zero jumb/'c2pa' byte-scan on the 480×599 variant is Hlias's and stays his, not admitted here.) No APP11/JUMBF segment ⇒ no in-file C2PA manifest. ADMITTED by Piotr after a substantive change — final characterization confirmed by Piotr; FactNotebook incorporated the revised wording (admission is Piotr's, not FN's)
on a SEPARATE ChatGPT image (posted 2026-10-01, not the case-3 post), the LinkedIn 'cr' badge DISPLAYS manifest data (OpenAI Media Service API; OpenAI OpCo, LLC; issued 2026-10-01)	**ESTABLISHED**	Piotr opened the 'cr' badge 2026-10-01 on a different image — directly observed display (same Polish LinkedIn UI, same labels, issued 2026-10-01). ONLY the display is established here: that image's served bytes were NOT inspected, so whether they lack the manifest is NOT established for this image. The displayed ≠ in-file conclusion rests on case 3 (H4-badge-displays-case3). ADMITTED by Piotr

on the case-3 post itself, the LinkedIn 'cr' badge DISPLAYS provenance values matching the case-3 manifest (app OpenAI Media Service API; publisher OpenAI OpCo, LLC; issued 2026-09-28)	**ESTABLISHED**	Piotr opened the case-3 post badge on 2026-10-01 (the served-bytes inspection was 2026-09-29): app OpenAI Media Service API, publisher OpenAI OpCo, LLC, issued 2026-09-28 — matching the case-3 manifest FIELDS (badge app = claim_generator 'OpenAI Media Service API'; badge publisher = signature_info.issuer 'OpenAI OpCo, LLC'; issued = signed date 2026-09-28). Viewed in the Polish LinkedIn UI: 'U■ywana aplikacja lub urz■dzenie: OpenAI Media Service API', 'Publikuj■cy: OpenAI OpCo, LLC', 'Wydano 28 wrze■nia 2026' (English rendering faithful). On the SAME image LinkedIn DISPLAYS the manifest's data while the served file carries no manifest — displayed ≠ in-file, on case 3 itself. ADMITTED by Piotr
the case-3 'cr' badge reads the displayed values from a C2PA manifest embedded in the served JPEG	**CONTRADICTED**	CONTRADICTED within the scope of the inspected files: the served bytes carry no manifest (H4-served-carries-manifest), so there is no embedded manifest in the served JPEG for the badge to read from

Limitations

H1-zimage-in-manifest — *the case-2 SIGNED manifest establishes the generating model as 'Z-Image'* → **CANNOT ESTABLISH** (no_channel). CANNOT ESTABLISH — 'Z-Image' is NOT in the signed manifest (which names only fal.ai, no model); the model identity rests on the README / Higgsfield selection record, a channel distinct from the signed declaration. Reason class: evidence/relationship (asserted on a different channel from the one that is signed)

H1-openai-produced-established — *the signed manifest ESTABLISHES that OpenAI actually produced this image (declaration → truth)* → **CANNOT ESTABLISH** (no_channel). a manifest is a DECLARATION, graded as a declaration; the signing credential is untrusted under the run config (no trust list) and no independent production-record channel is admitted — declared ≠ established

H2-injection-actor — *Higgsfield is the actor that injected the hf-job-id chunk into the delivered file* → **CANNOT ESTABLISH** (no_channel). Piotr's INFERENCE from the chunk's name and value; README 'Not claimed' — Higgsfield has not confirmed it. Reason class: evidence/relationship (a confirmation could resolve it)

H2-thumbnail-provenance — *the provenance of the delivered thumbnail (webp) can be established from the delivered bytes* → **CANNOT ESTABLISH** (no_channel). c2patool on 3_thumbnail_as_delivered.webp: 'No claim found' — a distinct representation with no provenance channel. Reason class: evidence/relationship (no channel

on this artefact)

H3-intrinsic-valid — *the case-3 export is valid as an INSTRUMENT-INDEPENDENT property of the file* → **CANNOT ESTABLISH** (no_channel). CANNOT ESTABLISH — reason class: INSTRUMENT / under-characterisation. The verdict is a function of validator version and trust configuration, not of the bytes alone (see H3-verdict-version-dependent). Pinning the validator version + config does NOT establish intrinsic validity — it establishes a DIFFERENT, instrument-relative proposition ('valid under validator X, config Y'); instrument-independent validity stays CANNOT ESTABLISH

H4-content-match — *the LinkedIn-served 480×599 JPEG is the same image as the case-3 export, downscaled (content match)* → **CANNOT ESTABLISH** (no_channel). CANNOT ESTABLISH in this admitted reconstruction — the only support was Hlias Staurou's PSNR measurement (served 480×599 vs v2 export = 32 dB vs ~8 dB controls), which Piotr did not run and did not admit on Hlias's behalf. It remains a DECLARED EXTERNAL RESULT attributed to Hlias; with no admitted observation, the content-match is not established here. (It was ESTABLISHED in the FN-admitted version FN-ce30f1089141d0fa; the authority's decision changes the state — this is No Exempt Node, demonstrated not declared.)

H4-served-from-exact-bytes — *the served JPEG was produced from those exact signed bytes* → **CANNOT ESTABLISH** (no_channel). the PSNR establishes content-match + manifest-absence, NOT byte-provenance: 'same image downscaled' ≠ 'came from those exact bytes'. Reason class: evidence/relationship

H4-badge-source-unknown — *the source the case-3 badge reads these displayed values from is established* → **CANNOT ESTABLISH** (no_channel). that the badge SHOWS these values is observed (H4-badge-displays-case3); reading from an embedded manifest in the served JPEG is excluded (H4-badge-reads-embedded CONTRADICTED); the ACTUAL source it uses (a LinkedIn-side store, the original upload, …) remains unobserved. Reason class: evidence/relationship

Separation of authority

Observation instruments determine what they observe and propose. Admission determines what may enter the reconstruction record. FactNotebook determines what that admitted record permits us to establish about the proposition. An instrument may search, select, transform or directly acquire observations according to its own declared procedure; those operations remain attributable properties of the instrument, not hidden properties of the reconstruction engine.

recorded behaviour → **observation** (attributed to its instrument) · declared risk / rule classifications → **the evaluator's / authority's policy** (never engine facts) · the consequential judgment → **the accountable authority**, outside this reconstruction.

Re-derivability

Two levels are preserved. **The finding** re-derives: re-running the reconstruction against the admitted observations reproduces the reconstruction identifier `FN-785efaab56d2e1f7` and the states in this document — with no re-run of the probabilistic extraction. **The admitted record's boundary** is separately inspectable via the candidate admission register (Annexes A/B) and the acquisition boundary (Annex C). *Version the instrument · trace the extraction · freeze the admitted record · replay the reconstruction.*

References

Each observation points to a citable source a third party can re-walk; the status records how it was verified.

C2PA fixture pack v2 (Piotr Reder, eucomplyhub) — 3 cases: case1 manifest names BFL/FLUX.2; case2 manifest names fal.ai with NO model ('Z-Image' is the README selection-record, not the signed manifest); case3 OpenAI ChatGPT export. Delivered PNGs Invalid (dataHash.mismatch) · chunk_removed data-hash matches · thumbnails No-claim · case3 Valid (no trust list) · *PRIMARY — supplied; SHA256SUMS re-verified here (7/7 OK)* · `supplied via iCloud 2026-09-30 (c2pa_fixtures_2026-09-30_v2)`

case3 c2patool output — validation_state Valid; claim generator 'OpenAI Media Service API'; signing cert subject CN 'OpenAI Media Service', issuer 'OpenAI OpCo, LLC'; digitalSourceType trainedAlgorithmicMedia; signed 2026-09-28T15:40:40Z; SHA-256 f564e3e0…b1f · *PRIMARY — in pack (c2patool 0.27.22, no trust list)* · `case3_chatgpt_linkedin/1_as_generated_chatgpt.png.c2patool.txt`

Validator version divergence (case 3) — case 3: c2pa-core 0.90.0 Valid; 0.91.0 Invalid 'certificate missing required EKU'; Valid again with an OpenAI EKU OID allow-listed. Which certificate (leaf/intermediate/TSA) 0.91 rejects on was NOT isolated;

no judgment on whether the 0.91 change is correct. Cases 1–2: Valid on both versions · *ADMITTED via Piotr's macOS reproduction (c2pa-python 0.37 & 0.38, core 0.90 & 0.91); discovery Hlias Staurou; not run by FactNotebook* · `LinkedIn thread (eucomplyhub)`

LinkedIn publication hop — served bytes (Piotr, 2026-09-29, 6 variants): progressive JPEG, JFIF only, no APP11/JUMBF, no OpenAI reference (Piotr recorded no 'c2pa' string search; the zero jumb/'c2pa' byte-scan on the 480×599 variant is Hlias's, not admitted). Badge (Piotr, Polish UI): case-3 post 'cr' shows OpenAI Media Service API / OpenAI OpCo, LLC / issued 2026-09-28 — matching the case-3 manifest, while the served file carries none; corroborated on a separate image (issued 2026-10-01). PSNR content-match (Hlias) = declared external result, NOT admitted → content-match CANNOT ESTABLISH · *ADMITTED by Piotr (served bytes + badge); Hlias's PSNR NOT admitted* · `https://www.linkedin.com/posts/piotrreder_c2pa-contentcredentials-aiact-share-7510335661502242817-UlVG/`

Annex B — Non-admitted candidate register

Candidates considered but not admitted. REJECTED = failed a criterion; UNRESOLVED = held pending adjudication (e.g. instruments diverge — Annex E); OUTSIDE ADMISSION SCOPE = potentially relevant, but this authority is not empowered to admit it (it enters only via another authority path, e.g. the originating party's direct confirmation). Record/admission statuses, not proposition truth-states. Recorded so exclusions remain inspectable, without contributing to reconstruction.

Candidate	Status	Relates to	Applied criterion	Reason
H-PSNR	**OUTSIDE-ADMISSION-SCOPE**	Hlias PSNR: served 480×599 vs v2 export = 32 dB (controls ~8 dB)	outside Piotr's admission authority (Hlias's measurement)	a declared external result attributed to Hlias; NOT held pending Piotr's decision — it enters only via Hlias's own authority path (direct confirmation). Its non-admission is why H4-content-match is CANNOT ESTABLISH
c2pa-string byte-scan	**OUTSIDE-ADMISSION-SCOPE**	Hlias byte-scan on the 480×599 variant	outside Piotr's admission authority (Hlias's measurement)	Hlias's scan; Piotr recorded no 'c2pa' string search, so it is kept out of OBS-C2PA-003 and enters only via Hlias's own authority path

Annex C — Acquisition / selection boundary

Records were supplied by an independent source (Piotr Reder (eucomplyhub) — public C2PA fixture pack v2). Completeness beyond the supplied surface is not established. The instrumentation was selective, not exhaustive: a selective extraction is not an exhaustive observation of the surface.

Surface constituted / submitted by	Piotr Reder (eucomplyhub) — public C2PA fixture pack v2
Submission form	supplied fixture set (3 cases, c2patool outputs, chunk_order, SHA256SUMS)
Candidate extraction instrument	c2patool 0.27.22 (Piotr's run, no trust list) — re-read here, not re-run
Admission authority / mechanism	Piotr Reder (eucomplyhub) — INDEPENDENT admission authority, within the records he can verify (admission register, 2026-10-02): he admitted/changed each interpretive candidate against his own records and marked anything outside that scope. FactNotebook retains extraction, traceability and execution. Direct records (FN's SHA-256 7/7; Piotr's preserved c2patool outputs) enter without a separate admission step. Hlias Staurou's measurements stay his: the 0.90/0.91 divergence entered via Piotr's own reproduction (discovery Hlias); the PSNR was NOT admitted (declared external result)
Reconstruction engine	camsva 0.9.0
Surface type	independently_supplied
Acquisition mode	selective

Known limitations	Bounded to the supplied pack. Pack primaries (files + SHA256SUMS) re-verified by FN (7/7 OK) → A2. c2patool VERDICTS are Piotr's runs, re-read not re-executed. Interpretive candidates were ADMITTED by Piotr within his verifiable records; Hlias's PSNR was not admitted, so the content-match is CANNOT ESTABLISH here (a declared external result). This establishes what the admitted record permits — not the underlying facts about who produced or altered the images. Decisions recorded in the accompanying admission register (Piotr Reder, 2026-10-02).

Annex D — Policy & engine binding

Engine camsva 0.9.0 · Reconstruction ID `FN-785efaab56d2e1f7`

FactNotebook reconstructs what can be established from the surviving records — including what cannot be — and keeps the establishment separate from the judgment applied to it.